

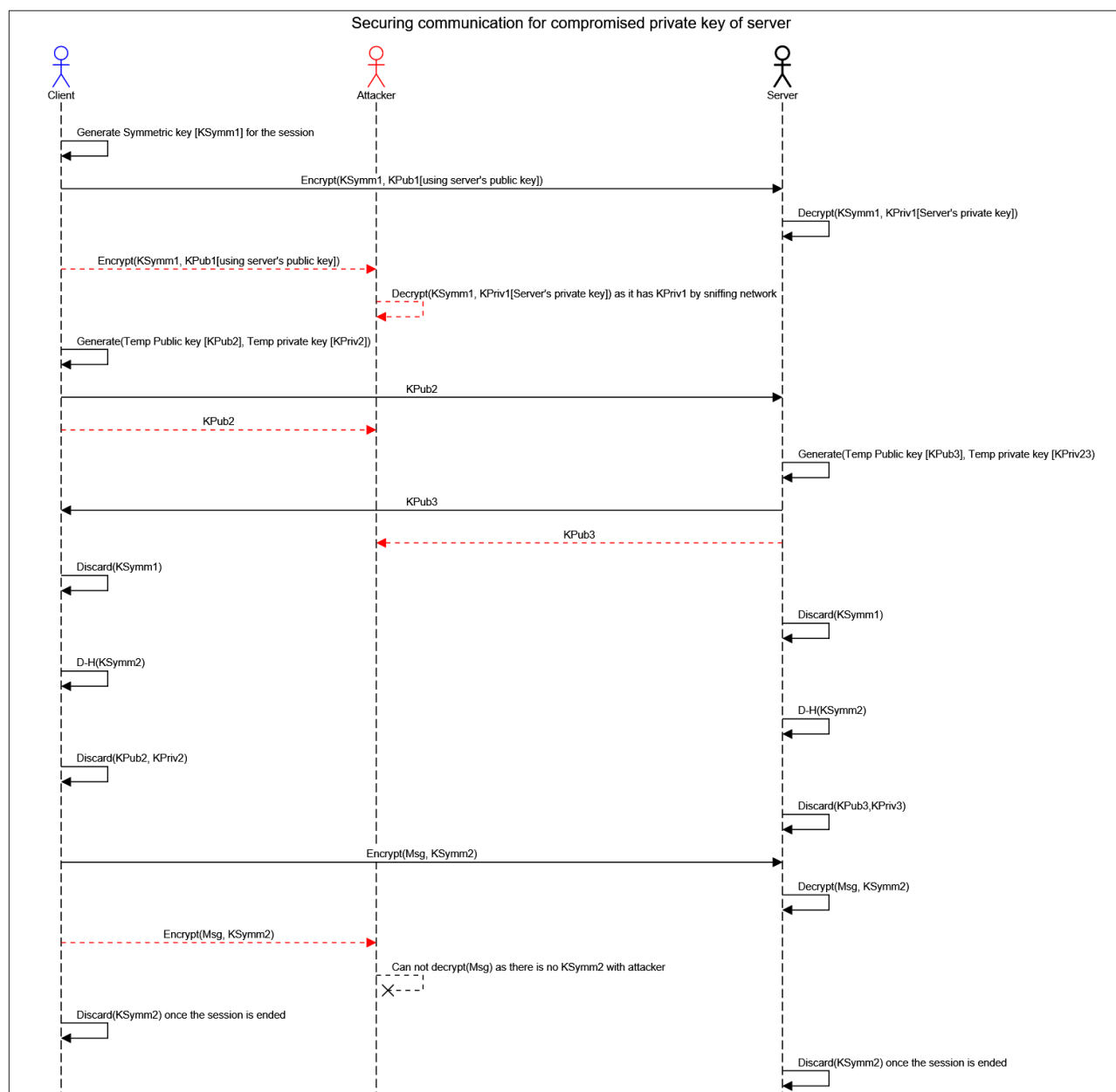


Figure 3: Mechanism using temporary keys to mitigate the compromise of private key. Note that communications with an attacker are highlighted in red

- Client sends the encrypted KPub2 to the server. Server extracts the public key KPub2 using KSymm1 it has. This encryption using KSymm1 is not essential as an attacker can get KSymm1 using the compromised private key of the server KPriv1.
- Server generates the private key and public key [KPub3 and KPriv3]. The server will then encrypt the KPub3 using KSymm1 as:
 - Encrypt(KPub3, KSymm1)
- Server sends this encrypted public key KPub3 to the client. The client will extract the KPub3 with KSymm1 it has.
- The key holding status for the client and the server is as follows:
 - Client has
 - KPub3 [server's temporary public key]
 - KPriv2 [client's own temporary private key]
 - Server has
 - KPriv3 [server's own temporary private key]
 - KPub2 [client's public key which is temporary]
- Now the client and server will both discard the KSymm1 after ensuring that server and client have their temporary asymmetric public keys exchanged.